

Long Term Animal Tracking - Hybrid Harvesting Concept

Long Term Animal Tracking System - Hybrid Harvesting Concept

Goal

Design an energy-autonomous tracking device for a sheep-size animal.

Main requirements:

- Position + activity datapoint at least every 1 min
- Local data logging, upload to base station every few days
- Maximum free-field transmission range: 100 m
- Lifetime: at least 2 years
- Total weight: below 1000 g
- Only commercially available components

Recommended Concept

Use a hybrid energy harvesting collar:

```
Solar      = main energy source
Kinetic    = motion-dependent supplement
Thermal    = continuous trickle source
Battery    = long-term buffer
LIC/supercap = pulse buffer for GNSS and LoRa
```

This is more robust than pure thermal or pure kinetic harvesting. It is also more distinctive than a simple solar-only tracker because it uses animal-specific energy sources.

System Architecture

```
Solar panel
-> solar PMIC / MPPT
-> storage

Kinetic generator
-> rectifier / harvester interface
-> storage

TEG thermal module
-> low-voltage boost converter
```

-> storage

Storage

-> regulators + load switches

-> MCU + GNSS + accelerometer + flash + LoRa

Operating Assumptions

- Datapoint interval: 1 min
- Each datapoint: timestamp, GNSS position, sleeping/awake state, storage voltage
- Upload interval: every 3 days
- Wireless link: LoRa to local base station
- GNSS and LoRa are power-gated when not used

Daily Energy Consumption

Module	Assumption	Energy
GNSS	621 mJ/fix, 1440 fixes/day	0.248 Wh/day
MCU	periodic low-power wake-up	0.005 Wh/day
Accelerometer	ADXL362, <2 uA	0.0002 Wh/day
Flash logging	one write/min	0.001 Wh/day
LoRa upload	batch upload every 3 days	0.006 Wh/day
PMIC/regulator losses	approx. 10-15% margin	0.030 Wh/day

Total:

0.291 Wh/day
= 1046 J/day

Daily Harvested Energy

Source	Assumption	Energy
Solar	2 W panel, 0.5 peak sun h/day, 70% efficiency	0.70 Wh/day
Thermal	100 uW average	0.0024 Wh/day
Kinetic	conservative animal motion estimate	0.00056 Wh/day

Total input:

0.703 Wh/day

Energy margin:

$$0.703 / 0.291 = 2.4$$

So the design can close the energy budget even under conservative winter solar assumptions.

Storage

Main storage:

$$\begin{aligned} &1 \times 18650 \text{ Li-ion cell} \\ &3.6 \text{ V} * 3500 \text{ mAh} = 12.6 \text{ Wh} \end{aligned}$$

Autonomy without harvesting:

$$12.6 \text{ Wh} / 0.291 \text{ Wh/day} = \text{approx. } 43 \text{ days}$$

Add a 30 F LIC or supercapacitor to handle GNSS and LoRa current peaks.

Component Candidates

Function	Candidate
Solar harvester	Voltaic Systems 2 W / 6 V flexible panel
Kinetic harvester	Kinetron MSG32 / mechanical generator system
Thermal harvester	TEC Microsystems 1MC06-048-15_TEG or Matrix thermal module
Solar PMIC	e-peas AEM10941
Thermal PMIC	Matrix MCRY12, LTC3108, or AEM20940
GNSS	u-blox MAX-M10S
Activity sensor	Analog Devices ADXL362
MCU	TI MSP430FR5969 or STM32L4
Wireless	Semtech SX1262 LoRa
Memory	SPI NOR flash, e.g. Winbond W25Q
Load switch	Vishay SIP32431

Weight Estimate

Part	Weight
2 W solar panel	60-100 g

Part	Weight
Kinetic generator	18-30 g
Thermal modules + heat spreader	60-100 g
18650 battery	45-50 g
LIC/supercap	20-40 g
PCB + electronics + antenna	30-60 g
Enclosure	100-200 g
Collar/mounting	150-250 g

Total estimate:

500–830 g

Below the 1000 g limit.

Possible Improvements

1. Kinetic Harvester as Activity Indicator

The kinetic generator output can also be used to estimate activity.

Possible features:

- pulse count per minute
- peak voltage
- harvested energy per minute
- time above voltage threshold

Example interpretation:

Kinetic signal	Activity
almost no pulses	sleeping/resting
low intermittent pulses	awake/grazing
frequent high pulses	walking/running

Still keep the accelerometer as the primary activity sensor. The kinetic signal is useful as a secondary passive indicator.

2. Adaptive GNSS Interval

Reduce GNSS energy by changing the fix interval:

State	GNSS strategy
sleeping/resting	reuse last position or fix every 10-30 min
grazing/slow movement	fix every 5 min
active/walking	fix every 1 min
low storage voltage	reduce GNSS frequency

3. Energy-Aware Scheduler

Use storage voltage to decide system behavior:

Storage level	Behavior
high	full GNSS + normal upload
medium	reduced GNSS interval
low	activity logging only, rare GNSS
critical	safe mode: RTC + wake-up only

Key Conclusion

Pure thermal + kinetic harvesting is not enough for one GNSS fix per minute. The hybrid solution is:

small solar panel + kinetic generator + thermal harvester + rechargeable storage

Estimated daily consumption:

0.291 Wh/day

Estimated daily harvested energy:

0.703 Wh/day

This gives an energy margin of about 2.4 while keeping the total weight below 1000 g.